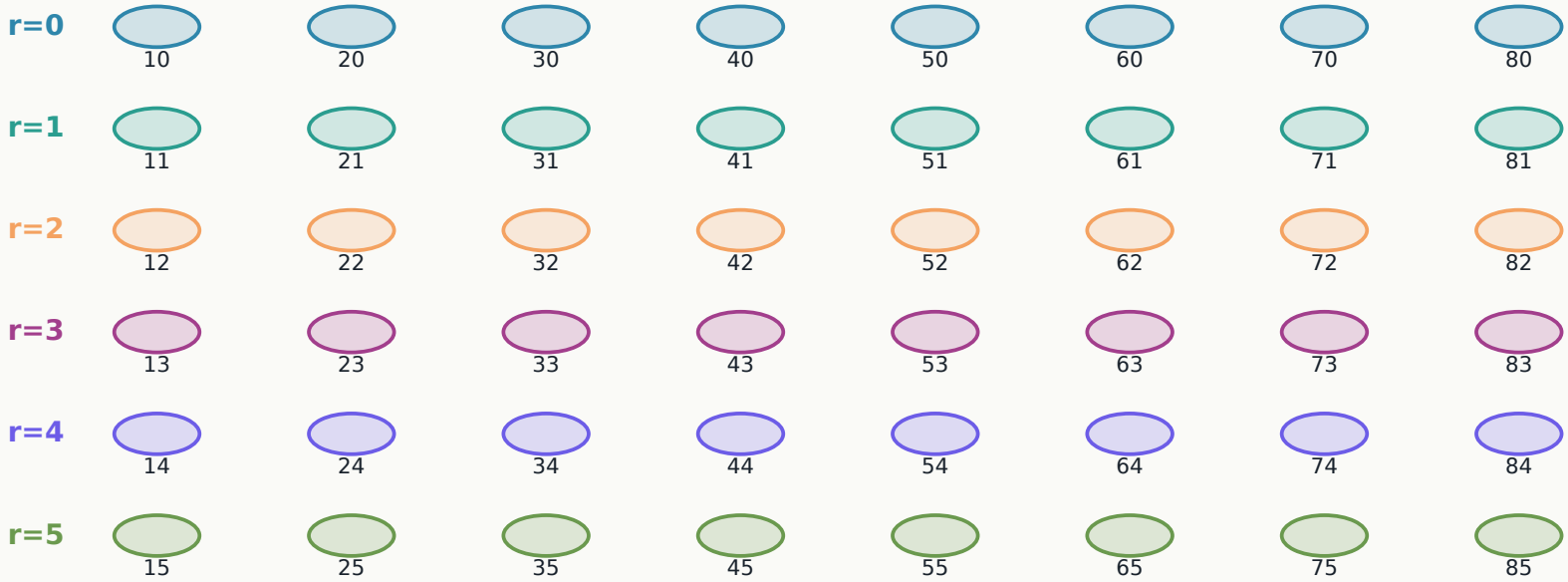
A high-frequency band is reorganized into offset polyphase channels



$$m_{qN+r}=\sum_j (a_jz_j^r)(z_j^N)^q+e_{qN+r}$$